Module Catalogue

Master of Science Robotics and Cognition

Examination regulations version 2024

Wintersemester 2026/2027 — as of 01.09.2026

Technische Hochschule Musterstadt

Structure of the programme

Area A — Compulsory modules: ROB-101, ROB-102

Area B — Elective modules: ROB-201

ROB-101 Robot Kinematics

Credits	6 ECTS
Weekly hours	4
Offered	every winter semester
Language	English
Module coordinator	Prof. Dr. Ada Beispiel
Learning outcomes	Students can derive the forward and inverse kinematics of serial manipulators.
Syllabus	• Homogeneous transformations • Denavit-Hartenberg convention • Jacobians and singularities
Assessment	Written exam, 90 minutes
Workload	60 h contact time, 120 h self-study, 180 h in total

ROB-102 Probabilistic State Estimation

Credits	6 ECTS
Offered	every summer semester
Language	English
Module coordinator	Prof. Dr. Karl Muster
Prerequisites	ROB-101 Robot Kinematics; linear algebra
Syllabus	• Bayes filter • Kalman filter and extended Kalman filter • Particle filters
Assessment	Oral exam, 30 minutes

ROB-201 Legged Locomotion

Credits	7.5 ECTS
Offered	irregularly
Language	German
Syllabus	• Zero moment point • Central pattern generators
Assessment	Project report